
accenture
AI reinvention made real
Accenture Innovation Challenge 2026

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UNDERSTANDING THE PROBLEM

Triage decides who waits, and in most emergency departments it is executed once, at the door. **Acuity is then treated as fixed while the patient's physiology keeps moving.** Three structural failures follow: judgment varies between nurses, half the room sits inside one crowded colour band with no ordering, and nothing re-measures a patient who was judged safe. A single adult-calibrated vital-sign scale then adds a fourth, because the same numbers mean opposite things at age 3 and age 78.

HOW IT FAILS - ONE PATIENT, NINETY-TWO MINUTES

T = 0
Arrives, triaged once

HR 92 · RR 18 · BP 128/80 · SpO_2 97%. No chest pain - all four inside the adult band.

ESI-4 · POSITION 11 / 14

T + 45
Still in the waiting room

No re-measurement is scheduled. No mechanism exists to look at her again.

UNOBSERVED

T + 75
Deteriorating, unseen

Diabetes blunts the cardiac pain that would have raised the alarm.

UNOBSERVED

T + 92
Critical event

Severe multi-vessel coronary disease. Ejection fraction 25-30%.

ESI-1 · TOO LATE

One decision. Ninety-two minutes. Nothing in the department was designed to revisit it.

THREE SEPARABLE GAPS - AND WHAT EACH ONE COSTS

GAP 01 · VARIABILITY

0.46-0.91

inter-rater agreement

Two competent nurses assign different levels to the same patient. Judgment, not protocol, decides.

OUR MECHANISM Deterministic acuity floor: four rule layers combined with min().

GAP 02 · QUEUE BLINDSPOT

50%+

of visits in one band

Inside a colour, arrival time decides position. A crowded ESI-3 band has no ordering at all.

OUR MECHANISM Continuous within-colour ranking by risk and trend.

GAP 03 · STATIC MEASUREMENT

92 min

unobserved, case P01

Triage executes once. Deterioration in the waiting room is invisible until someone walks past.

OUR MECHANISM Re-triage loop on a timer, plus two mandatory triggers.

WHY NOW - THE EVIDENCE

66% — sensitivity of human triage for patients who go on to need a life-saving intervention. One in three is not identified.
Sax et al., JAMA Netw Open 2023 · 5.3 mn encounters, 21 hospitals

50% — of arrivals have no prior record on file, so history cannot be assumed as an input.
Round 2 brief reference parameter

ESI v5 — extended vital-sign checks to low-acuity patients explicitly to prevent under-triage.
ESI Handbook v5 · AIIMS three-tier, prospectively validated

425,000 — de-identified ED stays with linked outcomes are now public, so training no longer needs a hospital partnership.
MIMIC-IV-ED, PhysioNet

Jan 2026 — FDA CDS guidance: the clinician must be able to independently review the basis of a recommendation.
FDA CDS final guidance · DPDP Act 2023 names the processor boundary

FRAMING DECISIONS, TAKEN UP FRONT

Jurisdiction	**India · DPDP Act 2023**
Acuity scale	**ESI, five levels**
Display	**RED / YELLOW / GREEN**
Deployment	**100-500+ visits per day**
Evidence	**500 synthetic encounters**

WHAT WE ASSUME — AND LABEL AS OURS

Synthetic evidence. All engine figures are measured on data we generated against ground truth we assigned.

Crowding share, 50%. The only invented input in the top-down sizing chain.

Two-thirds haircut. Applied to our own simulated impact before we quote it.

ONE ADULT BAND, TWO OPPOSITE FAILURES

3-YEAR-OLD, FEBRILE

HR 140 · RR 28 · 38.5 °C · alert, capillary refill under 2 s

ADULT-CALIBRATED MODEL RETURNS

NEWS2 7 - emergency response

HR 140 is band-normal for ages 1-4; heart rate rises about 10 bpm per 1 °C of fever. Every observed cue is reassuring.

ERROR: OVER-TRIAGE → ESI-3 + fever pathway

78-YEAR-OLD, “NORMAL”

HR 95 · RR 18 · 37.0 °C · BP 112/68 · new confusion

ADULT-CALIBRATED MODEL RETURNS

Every parameter scores 0 - GREEN

Blunted fever response. Beta-blockade caps the tachycardia: HR 95 here is the equivalent of 130. Systolic 112 against a baseline of 150 is a 25% drop.

ERROR: UNDER-TRIAGE → ESI-2 + sepsis screen

ONE SET OF OBSERVATIONS, FOUR ANSWERS

SAME NUMBERS, FIVE AGE BANDS — the calibration thesis

HR 92 · RR 18 · SBP 128 · SpO_2 97 · 36.9 °C — held constant, scored in each band.

IF THIS PATIENT WERE	LEVEL	EWS
Infant · 6 mo	ESI-2	5
Child · 3 y	ESI-3	4
Child · 8 y	ESI-4	1
Adult · 34 y ·	ESI-5	0
Geriatric · 78 y	ESI-2	0

One set of observations, 4 different answers. A single adult-calibrated scale would return one.

`Computed by the JS mirror — hypothetical ages, no measured record.`

Live from the prototype: the same observations held constant and scored in each age band. A single adult-calibrated scale would return one answer for all five.

WHAT THE CONTEXT LAYER IS WORTH

HIGH-ACUITY MISSED · 500-ENCOUNTER COHORT

Adult-calibrated vitals only	45.0%
Risk modifiers removed	20.0%
Full engine as proposed	**7.5%**

Six times fewer misses at the same over-triage cost.

Over-triage held at 33.2% in the vitals-only baseline and 32.0% in the full engine - the improvement is not bought with extra false alarms. Removing the context layer nearly triples misses, so the layer is not a refinement; it is the product.

THE MECHANISM WE BUILT INSTEAD

Five vital-sign bands. 0-1, 1-4, 5-11, 12-17 and adult, plus a ≥ 65 profile working on deviation from the patient’s own baseline.

Deviation, not raw values. Features are engineered as distance outside the patient’s own age band - which is what makes the monotonic constraint clinically meaningful.

Observed cues are first-class. ACVPU, work of breathing, colour and gait are structured inputs. New confusion alone scores +3 on NEWS2; a vitals-only pipeline drops it.

WHY THE OLD PATTERN PERSISTS

Diabetes blunts cardiac pain. Up to 40% of infarctions in diabetics present atypically - the alarm signal is absent, not weak.

Older women present atypically. Nearly half of over-65s with an infarction arrive with no chest pain at all.

The benign pattern is more available. GI symptoms read as gastritis under time pressure, because gastritis is far more common.

MEASURED, NOT ASSERTED

16 of 21
hidden-sick caught
genuinely high-acuity on entirely normal vitals

0
monotonicity violations
8 features × 1,200 held-out rows

0.943
held-out AUROC
AUPRC 0.838 · Brier 0.058 · logistic baseline 0.934

3.0%
encounters abstain
confidence below 0.50 routes to a clinician

SYNTHETIC · NOT CLINICAL

Ground truth is constructed at generation, independent of the engine — these figures show the architecture behaves as designed, not that it helps patients.

THE DETERMINISTIC ACUITY FLOOR

MACHINE LEARNING
Monotonic GBM finds hidden risk the rules cannot see.

MAY RAISE ACUITY — ESI-4 → ESI-2, without asking permission

DETERMINISTIC ACUITY FLOOR — min ( )
ESI Point A · red-flag rules · age & history modifiers · NEWS2 five bands

NO PATH DOWN EXISTS IN CODE — No component can lower a level

LICENSED CLINICIAN — THE ONLY DOWNWARD PATH
Reason code plus re-measured evidence, under their licence number.

THE INVARIANT · MAY RAISE, NEVER LOWER

The system is allowed to be cautious. It is not allowed to be falsely reassuring and that is a property of the code, not of the training run.

WHEN THE ENGINE DOES NOT KNOW

Missing data is not permission to guess. Below 0.50 confidence the engine abstains, holds the level rather than lowering it, and routes the case to a clinician - naming the test that would resolve the uncertainty, never a diagnosis.

0.47	2.8%	11.4%
confidence, case P14	missed · with a record	missed · no record

WHY THE GUARANTEE HOLDS

Enforced by construction

The four rule layers combine with min(); the learned score is clamped to the result. It is a constraint, not a weighting we hope holds.

Provable without a test set

Monotonicity is a property of the code, so the guarantee holds on data the model has never seen — including the day the case mix shifts.

Verified · zero violations

Every clinical feature raised by 1 SD across 8 features × 1,200 held-out rows. Worse physiology never once scored safer.

TRAINED ON OUTCOMES, NOT OPINIONS

Train on the acuity a human assigned and you teach the model to reproduce the miss. P01 was assigned low acuity by a competent nurse.

Labels come from what happened next - ICU transfer, admission, death, or intervention within 24 hours.

WHAT WE WILL NOT CLAIM

- Synthetic training proves the pipeline, not the medicine.
- Real data will need recalibration - Boston 2011-2019 in Uttar Pradesh is distribution shift.
- Subgroups are audited before deployment: 0.941 paediatric, 0.945 adult, 0.935 geriatric.
- No claim of clinical efficacy is made anywhere in this deck.

WHAT THE ARCHITECTURE INHERITS

Acuity scale	ESI v5, four decision points
Local protocol	AIIMS three-tier, validated
Vital scoring	NEWS2, five age bands
Training data	MIMIC-IV-ED, ~425,000 stays
Regulatory test	FDA CDS guidance, Jan 2026
Data law	DPDP Act 2023 · ABDM policy

SOLVED

Acuity scoring	solved
Safety architecture	solved
Deployability	solved
Clinical trust	not solved

WHY NOT THE CHEAPER ANSWERS

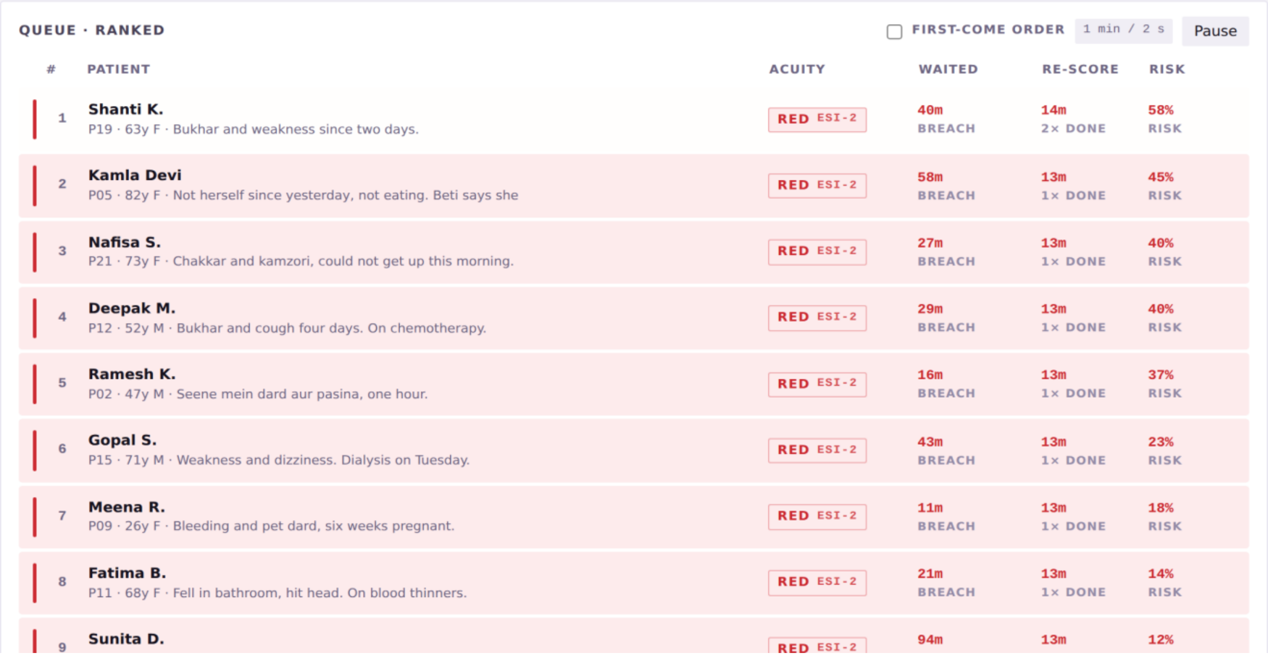
Do nothing. One in three critically ill patients is not identified, and the failure is invisible to the department it happens in.

Hire two more triage nurses. Buys throughput, not consistency - two more sets of judgment ranging 0.46-0.91, and none of them re-checks the waiting room.

Buy a commercial AI triage product. The canonical deployed example scored AUC 0.63 externally, missed 67% of sepsis, and alerted on 18% of all patients.

Every constraint in this deck derives from a published failure of a shipped system.

THE WAITING ROOM - RANKED, AND RE-SCORED ON A TIMER



1 Rank + movement

The ▲ badge shows how far a patient moved since the last re-score, and why.

2 Acuity, never colour alone

RED / YELLOW / GREEN always ships with the ESI level as text.

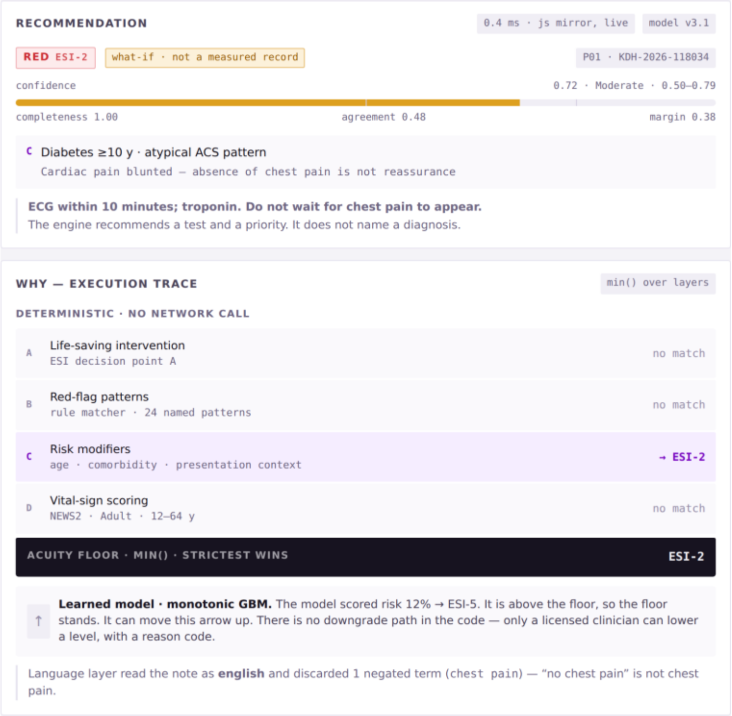
3 Safe-wait breach

Waited time turns red when it passes the safe window for that level.

4 Re-score counter

Every patient carries how many times the timer has re-evaluated them.

WHY - THE EXECUTION TRACE BEHIND EVERY SCORE



Every recommendation opens its own reasoning: which deterministic layers fired, the floor min() produced, the confidence figure and its band, and the note that the learned model may move the arrow up but has no downgrade path. No score is ever returned without a confidence figure - enforced as a failing test, not a convention.

Shanti K. arrived stable and moved thirteen places to #1 on a re-score - no nurse had to walk past her.

THE CLINICIAN'S JOURNEY, END TO END

1 · INTAKE

Vitals, free text in English or Hinglish, history and observed cues at the desk. Three taps, no typing on shift.

2 · SCORE

Four rule layers run deterministically on-device in under 800 ms. No network call in the clinical path.

3 · EXPLAIN

RED / YELLOW / GREEN with an ESI level, a confidence figure & at most three plain reasons or an explicit abstention.

4 · DECIDE

Accept with one tap, or override with one tap and a pre-populated reason code. Both cost the same.

5 · WATCH

The whole waiting room is re-scored on a timer and on every new observation; ranking updates itself.

6 · PROVE

Every decision and override writes an append-only, hash-chained record with model and ruleset version.

SYSTEM ARCHITECTURE - INPUT → INTERFACE → INTELLIGENCE → DATA → OUTPUT

USER / INPUT	INTERFACE	INTELLIGENCE	DATA / GOVERNANCE	OUTPUT / ACTION
Measured vitals available even on paper - highest weight	**Triage console** intake, score and disposition for one patient	**01 Life-saving check** ESI decision point A	**Tokenised at ingest** HMAC-SHA256(ABHA ‖ salt), salt in hospital HSM	**RED / YELLOW / GREEN** with ESI level, confidence and three reasons
Observed cues gait, speech, colour, work of breathing	**Waiting room** live ranked queue with breach countdowns	**02 Red-flag patterns** 24 named deterministic patterns	**On-premise** default-deny egress, air-gap capable	**Or an abstention** below 0.50 it routes to a clinician instead
Self-reported free text, English or Hinglish	**Three taps** card → accept or override → reason code	**03 Risk modifiers** age, comorbidity, presentation context	**Append-only log** hash-chained, WORM-backed, versioned	**Queue re-rank** continuous, within colour, by risk and trend
Record history richest, present for half of arrivals	**P95 < 800 ms** rules 15 · inference 40 · render 120	**04 Vital scoring** NEWS2 across five age bands	**Data Processor** hospital is Fiduciary under DPDP Act 2023	**Audit + exports** compliance packs, FHIR R4, delta feeds

The learned monotonic GBM sits above the floor and is clamped to it - it may raise the level, and has no code path to lower one.

ACUITY FLOOR = min(01…04)

A licensed clinician is the only downward path, with a reason code and re-measured evidence.

DEPLOYS AT THREE MATURITY TIERS

TIER 0 · PAPER ED ~100 visits/day

No EHR, manual vitals, intermittent power. Standalone offline tablet with CSV export - the deterministic layers and the re-triage timer need no network at all.

TIER 1 · DISTRICT ~250 visits/day

Basic HIS. One-way HL7 v2 ADT feed for registration and demographics; vitals still entered by hand.

TIER 2 · TERTIARY 500+ visits/day

Full EMR. Bidirectional FHIR R4 - Observation and Condition in, triage note back, Location for bed state.

No hospital needs Tier 2 to get the Tier 0 safety benefit.

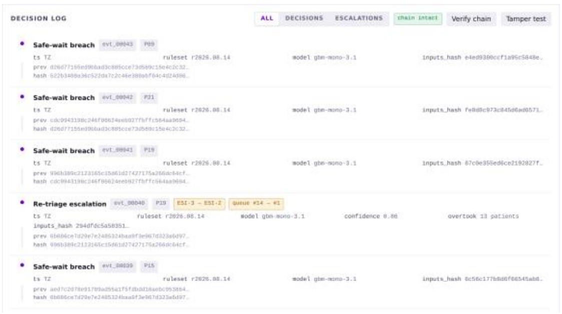
CONFIGURATION, NOT A CODE FORK

`threshold_profile` age-band table	`acuity_scale` ESI-5 or local
`wait_time_thresholds` safe windows	`surge_trigger` arrivals per hour
`language_pack` Hinglish, regional	`specialty_routing_map` pathways
`integration_tier` 0, 1 or 2	`escalation_contacts` alert recipients

Local protocol owns every hard threshold. No globally copied table ships, and no site needs a code fork.

Runs on one tablet - no GPU, no cloud, no API latency.

THE DATA BOUNDARY, AND WHO OWNS IT



Identity never enters. ABHA and MRN are tokenised at the gateway; re-identification needs two key custodians.

Nothing leaves, and every entry is provable. On-premise, default-deny egress. Each record commits to the one before it, so a silent edit breaks verification everywhere after.

ONE DEPARTMENT, ONE YEAR

109,500
annual visits at 300/day

10,950
high-acuity · 10% share

3,723
missed today · human 34%

821
missed with the engine · 7.5%

967 additional high-acuity patients identified per year - a two-thirds haircut on the simulated 2,902, because simulation will not transfer intact.

Top-down cross-check: 1.40 bn → 119 mn ED registrations → 60 mn in crowded EDs (est. 50%, ours) → 6.5 mn under-triaged (11%) → 6.5 lakh time-critical at the conservative 10% floor. Halving our crowding assumption still leaves ~3.3 lakh.

TUNED INSIDE THE TRAUMA STANDARD

4.0%
under-triage
ACS-COT ceiling is 5%

32.0%
over-triage
ACS-COT band is 25-35%

CONFUSION MATRIX · 500 SYNTHETIC ENCOUNTERS

	True 1-2	True 3	True 4-5
Pred 1-2	74	45	10
Pred 3	5	167	72
Pred 4-5	1	14	112

Amber cells are under-triage - the only errors that can kill. 20 of 500. Roughly eight over-triages are accepted to prevent one miss: one over-triage costs an avoidable ECG and about twelve minutes of nurse time; one under-triage costs a missed infarction. There is no exchange rate, which is why we never optimise average accuracy.

SYNTHETIC · NOT CLINICAL

THE HOSPITAL BUSINESS CASE

≈ ₹127 lakh — **modelled ICU delay avoidance a year**

35% of the 967 detections × 1.5 ICU days × ₹25,000. Sensitivity ₹85 L - ₹178 L across ₹18k - ₹35k per ICU day.

Flat — **over-triage cost**

32.0% against a 33.2% adult-calibrated baseline - the return is not funded by extra unnecessary workup.

≈ ₹19 lakh — **ward capacity released**

0.4 days average LOS reduction ≈ 387 bed-days; a further ₹19 lakh modelled from a 0.5 pp LWBS improvement.

₹35,000 — **Tier 0 deployment**

One tablet plus configuration. No integration project, and no licence in Phase 0 or Phase 1.

MODELLED · NOT MEASURED

Modelled payback under 12 months at Tier 2.

FIVE USERS, ONE MEASURE

Triage nurse
A ranked recommendation, three plain reasons, one tap.
Used, not worked around

Charge nurse
Queue depth, breach countdowns, one batched alert channel.
Zero un-actioned breaches

ED physician
The reasoning trace and the authority to move a patient down.
Override rate in a healthy band

Quality / NABH
Hash-chained audit trail, under-triage reporting by subgroup.
Tamper-evident evidence

Hospital CIO
A tier matching actual maturity; no PHI leaves the network.
Live without an IT project

Not the patient · not a diagnostic engine · not a staffing substitute - it changes the order of the queue, not its length.

AND IT HOLDS UNDER 3× LOAD

MEDIAN WAIT, HIGH-ACUITY PATIENTS · IDENTICAL ARRIVAL STREAMS

First-come order — 200.4

Continuously ranked — **8.4**

Minutes. Mean wait is unchanged by construction - ranking moves the sickest forward, so someone less sick waits longer. That is the trade, and it is the correct one. Every operational number degrades honestly: peak queue depth multiplies 22×, and 66% of safe-wait windows are breached at 3× volume.

No threshold, rule or floor moves under surge. Only how often we look.

AND THE ARGUMENT THAT SURVIVES A BOARD

Liability shield. Claims are lost on documentation, not only on care. Every decision writes what was known, what was recommended, who decided otherwise, and on what stated ground.

No new breach surface. Nothing identifiable leaves the network; no third party sits in a life-safety path.

Reversible. A charge nurse can switch it off without degrading the existing triage process.

NABH-ready. Append-only under-triage reporting by subgroup, with no extraction project.

The return is bought with detection, not with more false alarms.

PHASED ROADMAP - EVERY PHASE HAS A GATE, AND PHASE 0 IS A STOP

PHASE 0
months 0-3

Silent retrospective validation

The engine scores de-identified historical records. No clinician sees an output; no queue is reordered; no patient's care changes.

GATE · Under-triage below the department's own baseline

PHASE 1
months 4-6

Shadow mode, one ED

Runs live alongside clinicians with no autonomous impact on any patient's position in the queue.

GATE · Agreement rate and clinician trust

PHASE 2
months 7-12

Assisted mode under veto

Decision support inside the workflow. The clinician confirms every recommendation before it takes effect.

GATE · No increase in adverse events

PHASE 3
months 13-18

Multi-site, three tiers

Paper, district and tertiary departments on one engine - configuration only, no code forks.

GATE · Profiles work without code forks

PHASE 4
months 19-24

ABDM integration, scale

National identifier and consent architecture; sustained monitoring and per-site recalibration.

GATE · Sustained performance across sites

RISK AND MITIGATION

Risk	Control - architectural wherever possible	Owner
Patient harm from a wrong recommendation	No component can lower acuity; Phase 0 gate before any live use	Clinical advisor
Automation bias - the nurse stops thinking	Override rate monitored both ways; <2% or >30% triggers review	Charge nurse
Alert fatigue under surge	Alerts batch into one board; ≤ 6 alerts per nurse per hour	Charge nurse
Model drift after deployment	Monthly recalibration; every decision reproducible from the log	Quality lead
Distribution shift across sites	Rules floor is locally owned; recalibration is a Phase 3 gate	Local protocol
Device reclassification	Recommends a test and a priority, never a diagnosis; a human confirms	Regulatory
Data breach	On-premise; tokenised at ingest; every read audited	Hospital CIO
Clinician rejection of the tool	Shadow mode first; an override costs one tap and no form	Triage nurse

The strongest argument against us: every figure in this deck comes from data we generated, scored against ground truth we assigned. We accept it — which is exactly why the roadmap opens with a department's own historical records, and why the correct response to a failed Phase 0 is to stop, not to re-tune until it passes.

THE ASK - THREE INPUTS, NO CAPITAL

01 One partner emergency department
Any tier. A department running on paper is as useful to Phase 0 as a tertiary centre, because the engine needs no integration to be scored against.

02 De-identified historical records
Complaint, age, sex, recorded vitals, assigned acuity and what happened next: admission, ICU transfer, intervention within 24 hours.

03 One clinical advisor
An emergency physician or senior triage nurse to review the rule catalogue and hold local ownership of every hard threshold.

No integration | No capital outlay | No workflow change | No patient exposure

We are not asking a hospital to trust AI with its patients.

We are asking for the chance to prove whether it deserves that trust.

967 additional high-acuity patients a year, in one 300-visit department — the number we commit to.

Thank you